

Scenario D Explanation — Validation Case:

Scenario D demonstrates the input validation behavior of the simulator. Rather than running a scheduling simulation, it verifies that the application correctly rejects invalid process data before any computation takes place. The following invalid inputs are tested:

Invalid Input	Error Message Shown
Arrival Time = -3	Arrival Time cannot be negative.
Burst Time = 0	Burst Time must be greater than zero.
Burst Time = -5	Burst Time must be greater than zero.
Priority = "abc"	All numeric fields must be numbers.
Duplicate PID "P1"	Duplicate Process ID — each process must be unique.
Missing any field	All fields are required.
Priority = 0	Priority must be a positive integer (1–99).

Key Insight: The validation layer ensures that only clean, meaningful data reaches the scheduling engine. Each rule prevents a specific category of logical error: negative times are physically impossible, a zero burst time would cause division-by-zero in metric calculations, non-numeric priorities would break comparisons, duplicate PIDs would corrupt the process table, and missing fields would leave the algorithm with incomplete information. The screenshot above shows the live error message displayed beneath the input form when an invalid value is entered — the process is not added to the table until all fields pass validation.